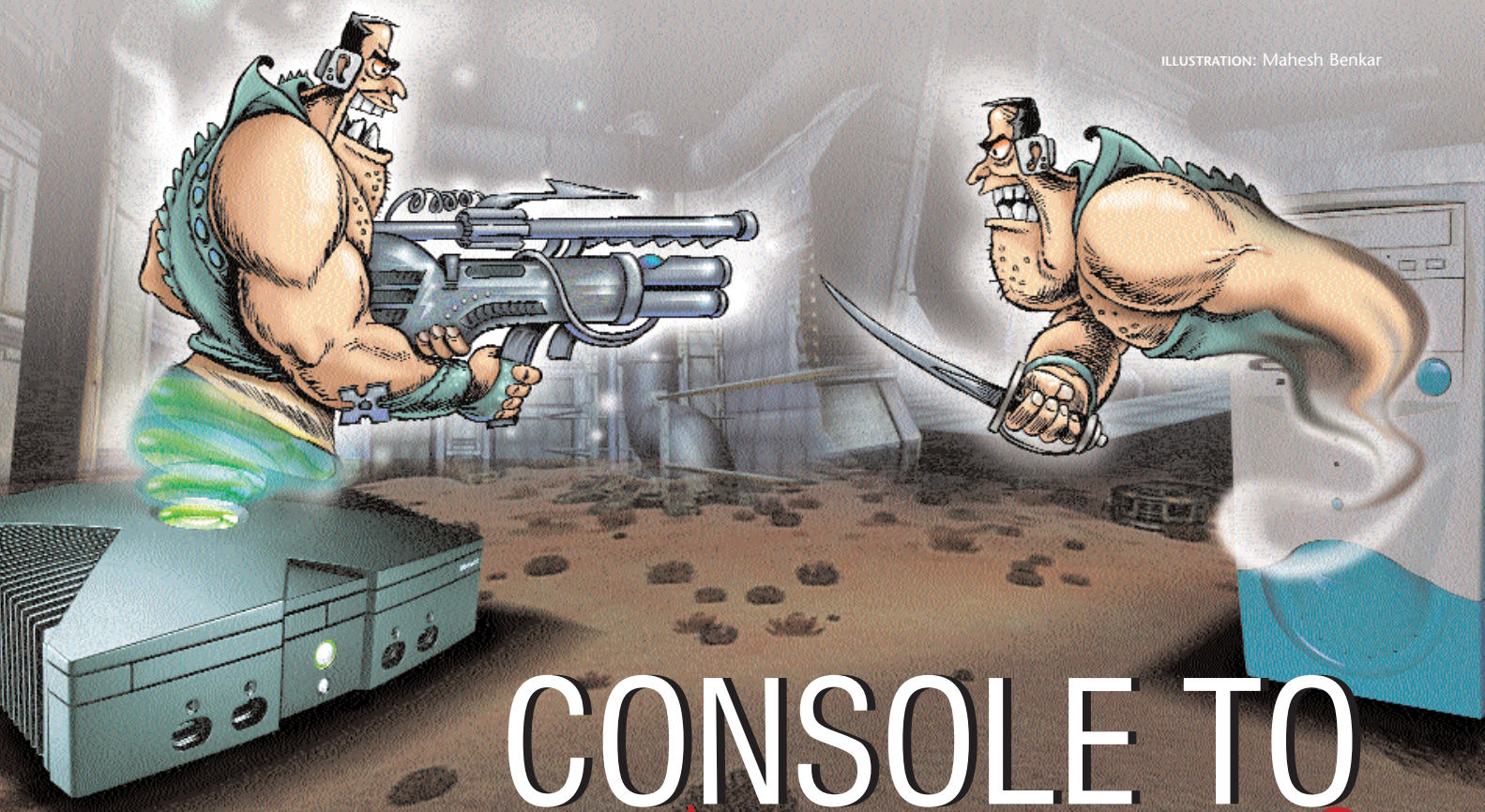




CONSOLE TO WIN!

Here's why the gaming console proves itself to be the ultimate consolation for the ever-growing demands of the diehard gamer.

Meet Jack and Jill—two hardcore gamers in town. Between them, Jack lives in the gaming world and plays his PS2. As for Jill, the poor thing is still deciding the best upgrade for her PC—a thing that is never on Jack's mind. Her father is on the verge of filing a lawsuit against a company that releases a new product every six months.

Agreed, the gaming console-versus-PC argument is getting pretty old but that's not the bone of contention here. Instead, by comparing the architectures of consoles and PCs, we can understand the difference between them. It will also help in understanding the requirements of not just games, but all 3D applications.

Design of the consoles

In the gaming arena, there is a sense of superiority when a gaming console is compared to a PC. It is difficult to believe that a gaming console that was designed

a couple of years ago, outperforms a PC that has a six-month product cycle. The fact remains that gaming consoles are better designed to handle the requirements of a gamer. It is actually a case of generalisation against specialisation.

The problem lies at the foundation of the design; the PC is designed for general purpose computing from the ground up. On the top there is some make up in the form of a GPU (sorry nVidia) and some buzzwords such as MMX, 3DNow, etc. Also, the PC platform is a perfect place for studying chaos theory. There are hundreds of manufacturers for the same number of components that make a PC. This throws up numerous restrictions—the video card manufacturer has to keep in mind the design of the motherboard, drivers may develop incompatibility with a specific component, etc. But a gaming console is developed for the sole purpose of playing games, and nothing else.

Another factor is that multimedia components have different processing requirements as compared to normal applications, such as office suites. In general-purpose software, different instructions are applied over localised data. In the case of a multimedia application, the reverse holds true. A set of instructions is repeatedly applied over large amounts of data—for example, the instructions for decoding an MP3 will be repeated over an entire file.

What do they actually do?

Good question. Let's look at the steps that are followed inside these boxes before they create those stunning visuals. In any 3D application, the first step is the creation of objects in a virtual 3D world. This step is called the geometry stage, where the objects and their positions are calculated. The output of this stage is what is called a 'display list'. This is a